

Program: 7

Implementation of K-Nearest Neighbor for any given dataset with k=7 and k=3.

Aim:

To implement of K-Nearest Neighbor for any given dataset with k=7 and k=3.

Algorithm:

Step1: Import required libraries.

Step2: Load and prepare the dataset.

Step3: Split into training and testing sets.

Step4: Apply KNN with k=3 and k=7.

Step5: Evaluate and compare the models.

Source Code:

Step 1: Import libraries

import pandas as pd

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

Step 2: Load dataset (using Iris dataset as an example)

iris = load_iris()

X = iris.data

y = iris.target

Step 3: Train-test split (80% training, 20% testing)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Step 4: KNN with k=3

knn3 = KNeighborsClassifier(n_neighbors=3)

knn3.fit(X_train, y_train)

y_pred3 = knn3.predict(X_test)

```

# Step 5: KNN with k=7

knn7 = KNeighborsClassifier(n_neighbors=7)

knn7.fit(X_train, y_train)

y_pred7 = knn7.predict(X_test)


# Step 6: Evaluate K=3

print("----- K = 3 -----")

print("Accuracy:", accuracy_score(y_test, y_pred3))

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred3))

print("Classification Report:\n", classification_report(y_test, y_pred3))


# Step 7: Evaluate K=7

print("\n----- K = 7 -----")

print("Accuracy:", accuracy_score(y_test, y_pred7))

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred7))

print("Classification Report:\n", classification_report(y_test, y_pred7))

```

Output:

----- K = 3 -----

Accuracy: 1.0

Confusion Matrix:

```
[[10 0 0]
```

```
[ 0 10 0]
```

```
[ 0 0 10]]
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	10
2	1.00	1.00	1.00	10

----- K = 7 -----

Accuracy: 1.0

Confusion Matrix:

[[10 0 0]

[0 10 0]

[0 0 10]]

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	10
2	1.00	1.00	1.00	10
